

Q19.

1) Overall Misclassification error = $1 - \max(P(+), P(-))$
 before split $= 1 - \frac{6}{10} = \frac{4}{10} = 0.4$
 $P(+) = \frac{4}{10}$, $P(-) = \frac{6}{10}$

Gain after

2) ~~Split on A~~ Split on A:

A	#	+	-
T	7	4	3
F	3	0	3

$A = T$
 error = $1 - \max\left(\frac{4}{7}, \frac{3}{7}\right) = 1 - \frac{4}{7} = \frac{3}{7}$

$A = F$
 error = $1 - \max\left(\frac{0}{3}, \frac{3}{3}\right) = 1 - 1 = 0$

Avg = $\frac{7}{10} \times \frac{3}{7} + \frac{3}{10} \times 0 = \frac{3}{10}$

Gain = $0.4 - 0.3 = 0.1$

3)

B	#	+	-
T	4	3	1
F	6	1	5

$B = T$
 error = $1 - \frac{3}{4} = 0.25$

$B = F$
 error = $1 - \frac{5}{6} = \frac{1}{6}$

Avg = $\frac{4}{10} \times \frac{1}{4} + \frac{6}{10} \times \frac{1}{6} = \frac{1}{10} + \frac{1}{10} = 0.2$

Gain = 0.2

4) Choose B higher Gain

5) Entropy & Gini ~~both~~ better more sensitive to changes